OCR Further Mathematics
Discrete Mathematics
Question Paper
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1(a) Sarah is having some work done on her garden.

The table below shows the activities involved, their durations and their immediate predecessors. These durations and immediate predecessors are known to be correct.

Activity	Immediate predecessors	Duration (hours)
A Clear site	–	4
B Mark out new design	A	1
C Buy materials, turf, plants and trees	–	3
D Lay paths	B, C	1
E Build patio	B, C	2
F Plant trees	D	1
G Lay turf	D, E	1
H Finish planting	F, G	1

(i) Use a suitable model to determine the following.

- The minimum time in which the work can be completed
- The activities with zero float

[6]

Minimum time = Activities with zero float:

(ii) State one practical issue that could affect the minimum completion time in part (i).

[1]

(b) Sarah needs the work to be completed as quickly as possible. There will be at least one activity happening at all times, but it may not always be possible to do all the activities that are needed at the same time.

Determine the earliest and latest times at which building the patio (activity E) could start. [3]

(c) There needs to be a 2-hour break after laying the paths (activity D). During this time other activities that do not depend on activity D can still take place.

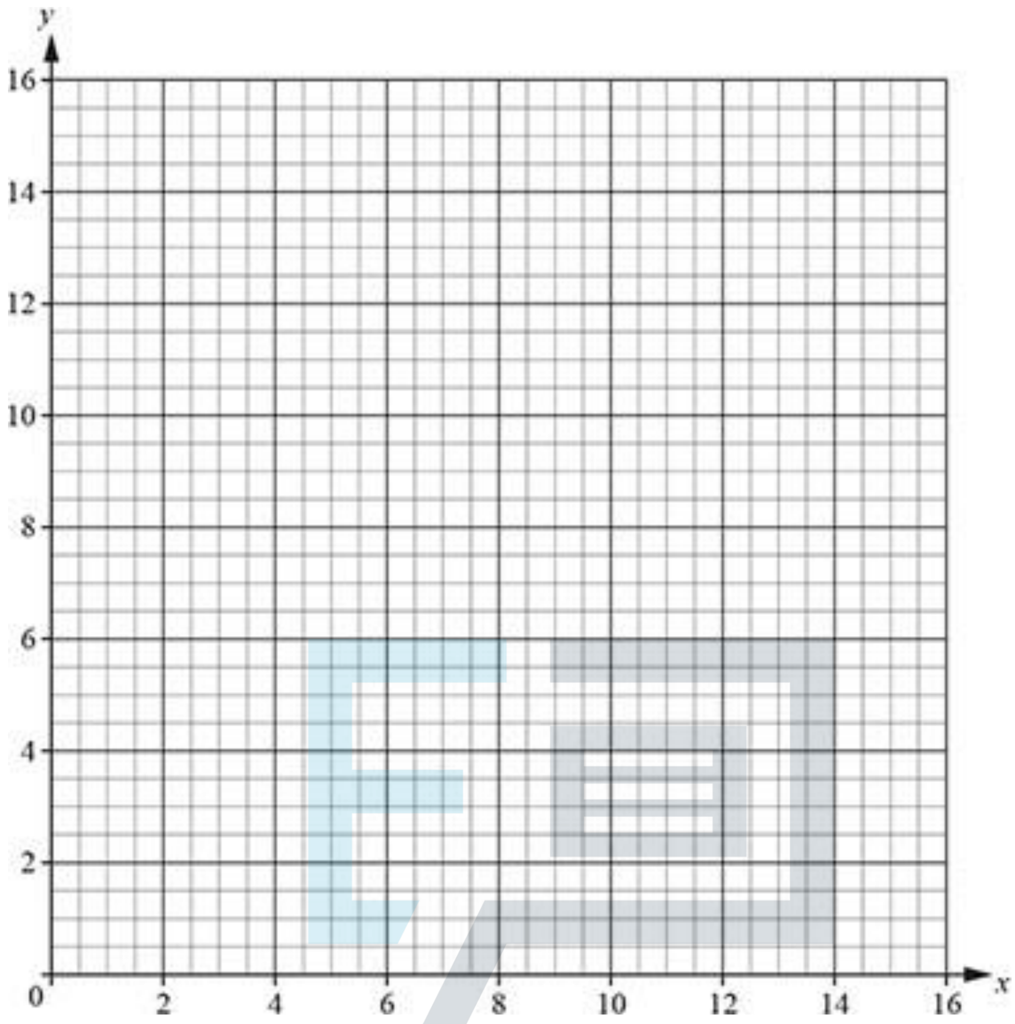
Describe how you would adapt your model to incorporate the 2-hour break. [1]

2(a) A linear programming problem is formulated as below.

Maximise $P = 4x - y$
subject to $2x + 3y \geq 12$
$x + y \leq 10$
$5x + 2y \leq 30$
$x \geq 0, y \geq 0$

(i) Identify the feasible region by representing the constraints graphically and shading the regions where the inequalities are not satisfied.

[4]



(ii) Hence determine the maximum value of the objective. [2]

(b) The constraint $x + y \leq 10$ is changed to $x + y \leq k$, the other constraints are unchanged.

Determine, algebraically, the value of k for which the maximum value of P is 3.

Do not draw on the graph above and do not use the spare grid. [3]

(c) Determine, algebraically, the other value of k for which there is a (non-optimal) vertex of the feasible region at which $P = 3$.

Do not draw on the graph above and do not use the spare grid. [2]

3(a) Li and Mia play a game in which they simultaneously play one of the strategies X, Y and Z.

The tables show the points won by each player for each combination of strategies. A negative entry means that the player loses that number of points.

Points won by Li		Mia X	Mia Y	Mia Z
Li	X	5	−6	0
	Y	−2	3	4
	Z	−1	4	8

Points won by Mia		Mia X	Mia Y	Mia Z
Li	X	4		
	Y	11		5
	Z	10	5	1

The game can be converted into a zero-sum game, this means that the total number of points won by Li and Mia is the same for each combination of strategies.

(i) Complete the table above to show the points won by Mia.

[1]

(ii) Convert the game into a zero-sum game, giving the pay-offs for Li.

[1]

Points won by Li		Mia X	Mia Y	Mia Z
Li	X			
	Y			
	Z			

(b) Use dominance to reduce the pay-off matrix for the game to a 2×2 table.

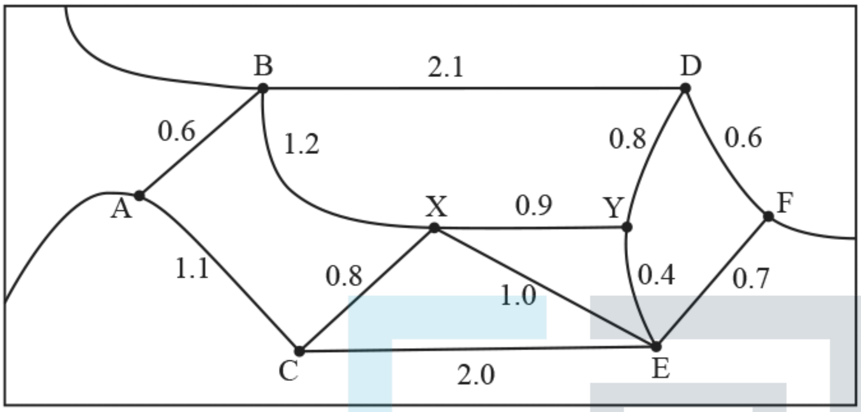
You do not need to explain the dominance. [2]

Points won by Li		Mia	
Li			

(c) Mia knows that Li will choose his play-safe strategy.

Determine which strategy Mia should choose to maximise her points. [3]

4(a) The diagram shows a simplified map of the main streets in a small town.



Some of the junctions have traffic lights, these junctions are labelled A to F.
There are no traffic lights at junctions X and Y.
The numbers show distances, in km, between junctions.

Alex needs to check that the traffic lights at junctions A to F are working correctly.

Find a route from A to E that has length 2.8 km. [1]

(b) Alex has started to construct a table of shortest distances between junctions A to F.

A					
	B				
	1.7	C			
	2.1	2.5	D		
	2.2	1.8	1.2	E	
					F

For example, the shortest route from C to B has length 1.7 km, the shortest route from C to D has length 2.5 km and the shortest route from C to E has length 1.8 km.

Complete the table above. [4]

(c) Use your table from the above part to construct a minimum spanning tree for the complete graph on the six vertices A to F.

- Write down the total length of the minimum spanning tree.
- List which arcs of the original network correspond to the arcs used in your minimum spanning tree.

[3]

Total length =

Arcs in the original network:

(d) Beth starts from junction B and travels through every junction, including X and Y. Her route has length 5.1 km.

Write down the junctions in the order that Beth visited them.

Do not draw on your answer from the above part. [2]

5(a) Seven items need to be packed into bins. Each bin has capacity 30 kg.
The sizes of the items, in kg, in the order that they are received, are as follows.

12 23 15 18 8 7 5

Find the packing that results using each of these algorithms.

(i) The next-fit method

[2]

Bin 1				
Bin 2				
Bin 3				
Bin 4				
Bin 5				

(ii) The first-fit method

[2]

Bin 1				
Bin 2				
Bin 3				
Bin 4				
Bin 5				

(iii) The first-fit decreasing method

[2]

Bin 1				
Bin 2				
Bin 3				
Bin 4				
Bin 5				

(b) A student claims that all three methods from above part can be used for both 'online' and 'offline' lists.

Explain why the student is wrong. [2]

(c) The bins of capacity 30 kg are replaced with bins of capacity M kg, where M is an integer.
The item of size 23 kg can be split into two items, of sizes x kg and $(23 - x)$ kg, where x may be any integer value you choose from 1 to 11. No other item can be split.

Determine the smallest value of M for which four bins are needed to pack these eight items. Explain your reasoning carefully.

[3]

6(a) A set consists of five distinct non-integer values, A, B, C, D and E.

The set is partitioned into non-empty subsets and there are at least two subsets in each partition.

Show that there are 15 different partitions into two subsets. [2]

(b) Show that there are 25 different partitions into three subsets. [4]

(c) Calculate the total number of different partitions. [2]

(d) The numbers 12, 24, 36, 48, 60, 72, 84 and 96 are marked on a number line. The number line is then cut into pieces by making cuts at A, B, C, D and E, where $0 < A < B < C < D < E < 100$.

Explain why there must be at least one piece with two or more of the numbers 12, 24, 36, 48, 60, 72, 84 and 96.

[2]

7(a) Tamsin is planning how to spend a day off. She will divide her time between walking the coast path, visiting a bird sanctuary and visiting the garden centre.

Tamsin has given a value to each hour spent doing each activity. She wants to decide how much time to spend on each activity to maximise the total value of the activities.

Activity	Walking coast path	Visiting bird sanctuary	Visiting garden centre
Value	5 points per hour	3 points per hour	2 points per hour

Tamsin's requirements are that she will spend:

- a total of exactly 6 hours on the three activities
- at most 3.5 hours walking the coast path
- at least as long at the bird sanctuary as at the garden centre
- at least 1 hour at the garden centre.

If Tamsin spends x hours walking the coast path and y hours visiting the bird sanctuary, how many hours does she spend visiting the garden centre?

[1]

(b) Tamsin models her problem as the linear programming formulation

Maximise $P = 3x + y$

Subject to

$x \le 3.5$

$x + 2y \ge 6$

$x + y \le 5$

and $x \ge 0$

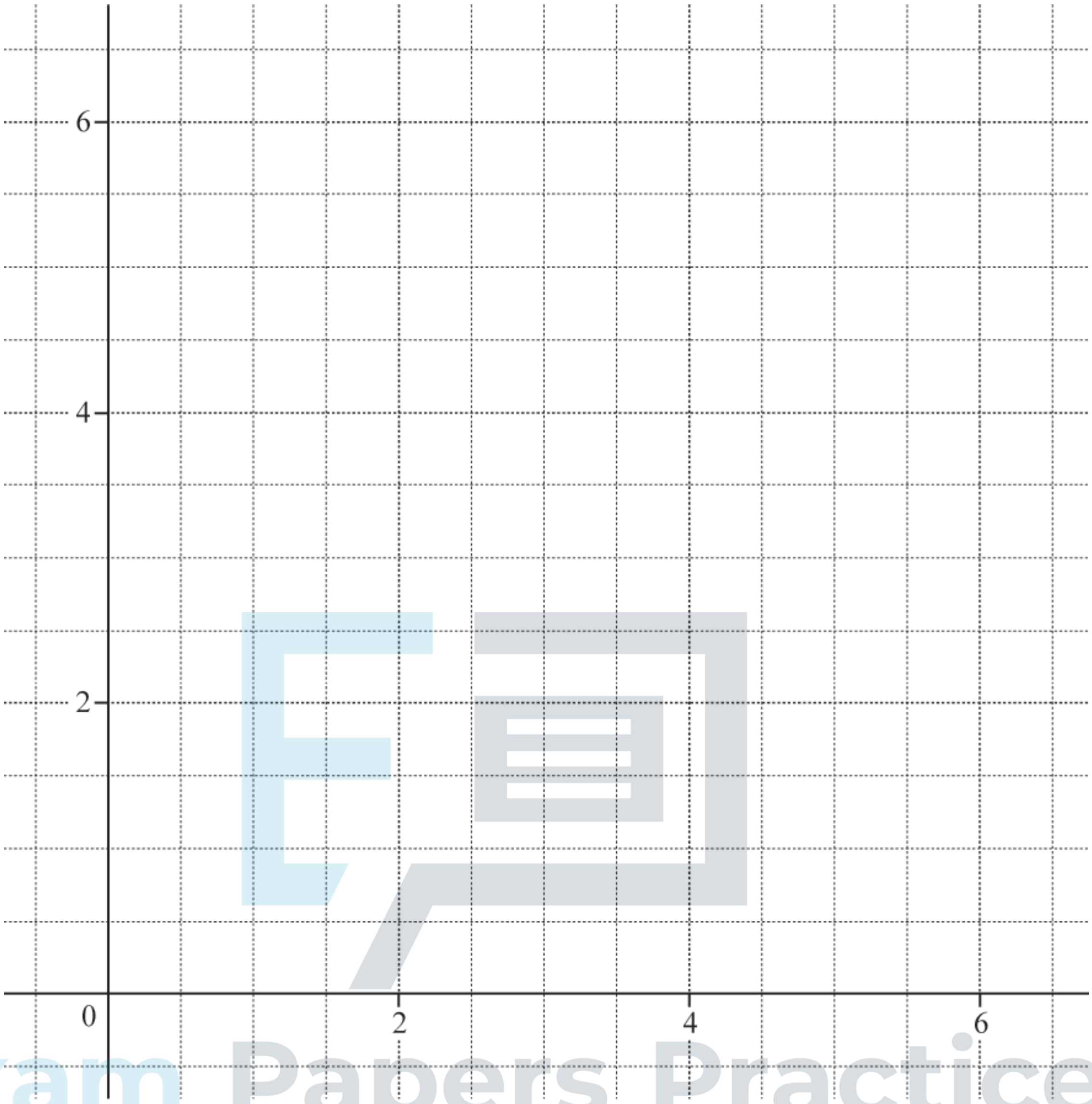
(i) Explain why the maximum total value of the activities done is achieved when $3x + y$ is maximised. [3]

(ii) Show how the requirement that she spends at least as long at the bird sanctuary as at the garden centre leads to the constraint $x + 2y \ge 6$.

[2]

(iii) Explain why there is no need to require that $y \geq 0$. [2]

(c) Represent the constraints graphically and hence find a solution to Tamsin's problem. [7]


0
2
4
6
2
4
6

8(a) The number of points won by player 1 in a zero-sum game is shown in the pay-off matrix below, where k is a constant.

Player 2

Player 1		Strategy E	Strategy F	Strategy G	Strategy H
	Strategy A	$2k$	-2	$1-k$	4
	Strategy B	-3	3	4	-5
	Strategy C	1	4	-4	2
	Strategy D	4	-2	-5	6

In one game, player 2 chooses strategy H.

Write down the greatest number of points that player 2 could win. [1]

(b) You are given that strategy A is a play-safe strategy for player 1.

Determine the range of possible values for k. [6]

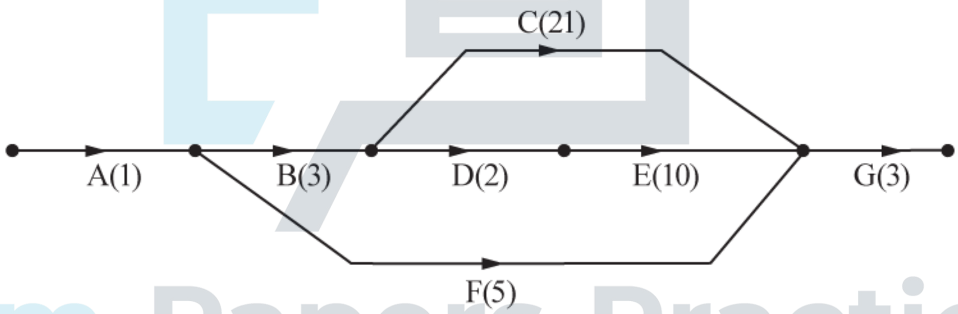
(c) Determine the column minimax value. [2]

9(a) Bob is extending his attic with the help of some friends, including his architect friend Archie.

The activities involved, their durations (in days) and Bob's notes are given below.

Activity		Duration (days)	Notes
A	Archie takes measurements	1	
B	Archie draws up plans	3	Must come after A
C	Plans are approved	21	Must come after B
D	Bob orders materials	2	Must come after B
E	Materials delivered	10	Must come after D
F	Work area cleared	5	Must come after A
G	Plumbing and electrics	3	Must come after C, E and F
H	Floors, walls and ceilings	24	Must come after G
I	Staircase	2	Must come after H
J	Windows	1	Must come after H
K	Decorating	6	Must come after I and J

Archie has started to construct an activity network to represent the project.



Complete the activity network given above and use it to determine

- the minimum completion time
- the critical activities

for the project. [6]

Minimum project completion time = days

Critical activities:

(b) Give two different reasons why the project might take longer than the minimum completion time. [2]

First reason:

Second reason:

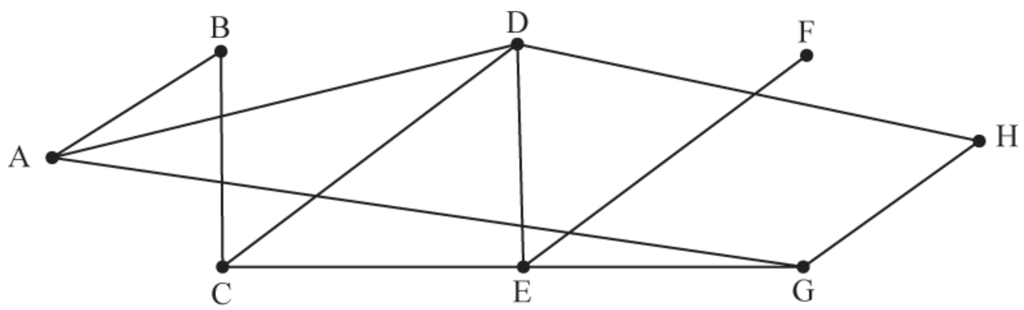
(c) For the activity with the longest float

- calculate this float, in days
- interpret this float in context. [2]

Longest float = days

Interpretation:

10(a)



By listing the vertices passed through, in the order they are used, give an example for the graph above of:

(i) a walk from A to H that is not a trail, [1]

(ii) a trail from A to H that is not a path, [1]

(iii) a path from A to H that is not a cycle. [1]

(b) The arcs are weighted to form a network. The arc weights are shown in the table, apart from the weight of arc AG. The arcs model a system of roads and the weights represent distances in km.

	A	B	C	D	E	F	G	H
A		3	–	8	–	–	x	–
B	3	–	5	–	–	–	–	–
C	–	5	–	4	3	–	–	–
D	8	–	4	–	5	–	–	9
E	–	–	3	5	–	7	6	–
F	–	–	–	–	7	–	–	–
G	x	–	–	–	6	–	–	2
H	–	–	–	9	–	–	2	–

The road modelled by arc AG is temporarily closed.

Use the matrix form of Prim's algorithm to find a minimum spanning tree, of total weight 30 km, that does not include arc AG. [3]

(c) The road modelled by arc AG is now reopened.

(i) For what set of values of x could AG be included in a minimum spanning tree for the full network? [1]

(ii) Which arc from the minimum spanning tree in part (b) is not used when arc AG is included? [1]

11(a) Jameela needs to store ten packages in boxes. She has a list showing the size of each package.
The boxes are all the same size and Jameela can use up to six of these boxes to store all the packages.

Which of the following is a question that Jameela could ask which leads to a construction problem? Justify your choice.

- In how many different ways can I fit the packages in the boxes?
- How can I fit the packages in the boxes? [1]

(b) The total volume of the packages is 1 m^3. The volume of each of the six boxes is 0.25 m^3.

Explain why a solution to the problem of storing all the packages in six boxes may not exist. [1]

(c) The volume of each package is given below.

Package	A	B	C	D	E	F	G	H	I	J
Volume (m^3)	0.20	0.05	0.15	0.25	0.04	0.03	0.02	0.02	0.12	0.12

By considering the five largest packages (A, C, D, I and J) first, explain what happens if Jameela tries to pack the 10 packages using only four boxes. [2]

(d) You may now assume that the packages will always fit in the boxes if there is enough volume.

Use first-fit to find a way of storing the packages in the boxes. Show the letters of the packages in each box, in the order that they are packed into that box. [2]

Box 1:

Box 2:

Box 3:

Box 4:

Box 5:

Box 6:

(e) The order of the packages within a box does not matter and the order of the boxes does not matter. So, for example, having A and E in box 1 is the same as having E and A in box 2, but different from having A in one box and E in a different box.

Suppose that packages A and B are not in the same box. In this case the following are true:

- there are 8 different ways to put any or none of E, F, G and H with A
- 3 include F with A
- 3 include G with A
- 1 includes both F and G with A

Use the inclusion-exclusion principle to determine how many of the 8 ways include neither package F nor package G. [2]

12(a) Algorithm X is given below:

STEP 1: Let $A = 0$
STEP 2: Input the value of B [B must be a positive integer]
STEP 3: C = INT($B \div 3$)
STEP 4: $D = B + 3C$
STEP 5: Replace A with $(D - A)$
STEP 6: Decrease B by 1
STEP 7: If $B > 0$ go to STEP 3
STEP 8: Display the value of ABS(A) and STOP

Trace through algorithm X when the input is $B = 5$.
Only record changes to the values of the variables, A, B, C and D, using a new column of the table given below each time one of these variables changes. [5]

A												

B												
C												
D												
A												
B												
C												
D												

Output =

(b) Explain carefully how you know that algorithm X is finite for all positive integer values of *B*. [2]

(c) Explain what it means to say that an algorithm is deterministic. [1]

(d) The run-time of algorithm X is k × (number of times that STEP 6 is used), for some positive constant k.

Explain, with reference to your answer to part (b), why the order of algorithm X is O(n), where n is the input value of B. [1]

(e) For each input B, algorithm Y achieves the same output as algorithm X, but the run-time of algorithm Y is m × (the number of digits in B), where m is a positive constant.

Show that algorithm Y is more efficient than algorithm X. [1]

13(a) Drew and Emma play a game in which they each choose a strategy and then use the tables below to determine the pay-off that each receives.

Drew's pay-off

		Emma X	Emma Y	Emma Z
Drew	P	3	14	11
Drew	Q	12	4	7
Drew	R	11	4	6

Emma's pay-off

		Emma X	Emma Y	Emma Z
Drew	P	13	2	5
Drew	Q	4	12	9
Drew	R	5	12	10

Convert the game into a zero-sum game, giving the pay-off matrix for Drew. [2]

		Emma X	Emma Y	Emma Z
Drew	P			
Drew	Q			
Drew	R			

(b) Determine the optimal mixed strategy for Drew. [5]

(c) Determine the optimal mixed strategy for Emma. [5]

14(a) Corey is training for a race that starts in 18 hours time. He splits his training between gym work, running and swimming.

- At most 8 hours can be spent on gym work.
- At least 4 hours must be spent running.
- The total time spent on gym work and swimming must not exceed the time spent running.

Corey thinks that time spent on gym work is worth 3 times the same time spent running or 2 times the same time spent swimming. Corey wants to maximise the worth of the training using this model.

Formulate a linear programming problem to represent Corey's problem.
Your formulation must include defining the variables that you are using. [5]

(b) Suppose that Corey spends the maximum of 8 hours on gym work.

(i) Use a graphical method to determine how long Corey should spend running and how long he should spend swimming. [5]

(ii) Describe why this solution is not practical. [1]

(iii) Describe how Corey could refine the LP model to make the solution more realistic. [1]

15(a) The table shows the activities involved in a project, their durations in hours and their immediate predecessors. The activities can be represented as an activity network.

Activity	A	B	C	D	E	F	G	H

Duration	2	4	5	4	3	3	2	4
Immediate predecessors	-	A	-	A, C	B, C	B, D	D, E	F, G

Use standard algorithms to find the activities that form

- the longest path(s)
- the shortest path(s)

through the activity network.

You must show working to demonstrate the use of the algorithms. [8]

Space for working:

Longest path(s):

Shortest path(s):

(b) Only one of the paths from the above part has a practical interpretation.

What is the practical interpretation of the total weight of that path? [1]

(c) The duration of activity E can be changed. No other durations change.

What is the smallest increase to the duration of E that will make activity E become part of a longest path through the network? [1]

16(a) Give an example of a standard sorting algorithm that can be used when some of the values are not known until after the sorting has been started. [1]

(b) Becky needs to sort a list of numbers into increasing order.
She uses the following algorithm:

STEP 1: Let *L* be the first value in the input list.
Write this as the first value in the output list and delete it from the input list.

STEP 2: If the input list is empty go to STEP 7.
Otherwise let *N* be the new first value in the input list and delete this value from the input list.

STEP 3: Compare *N* with *L*.

STEP 4: If *N* is less than or equal to *L*
- write the value of *N* immediately before *L* in the output list,
- replace *L* with the first value in the new output list,
- then go to STEP 2.

STEP 5: If *N* is greater than *L*
- if *L* is the value of the last number in the output list, go to STEP 6;
- otherwise, replace *L* with the next value in the output list and then go to STEP 3.

STEP 6: Write the value of *N* immediately after *L* in the output list.
Let *L* be the first value in the new output list and then go to STEP 2.

STEP 7: Print the output list and STOP.

Trace through Becky's algorithm when the input list is

6 9 5 7 6 4

Complete the table below, starting a new row each time that STEP 3 or STEP 7 is used. [5]

Input list	*L*	*N*	Output list

(c) Becky measures the efficiency of her sort by counting using the number of times that STEP 3 is used.

(i) How many times did Becky use STEP 3 in sorting the list from the part above? [1]

(ii) What is the greatest number of times that STEP 3 could be used in sorting a list of 6 values? [1]

(d) A computer takes 15 seconds to sort a list of 60 numbers using Becky's algorithm.

Approximately how long would you expect it to take the computer to sort a list of 300 numbers using the algorithm? [3]

17(a) Two graphs are shown below.

Graph G1

Graph G2

List the vertex degrees for each graph. [3]

Vertex	J	K	L	M	N
Degree					

Vertex	J	K	L	M	N
Degree					

(b) Prove that the graphs are non-isomorphic. [2]

(c) The two graphs are joined together by adding an arc connecting J and T.

(i) Explain how you know that the resulting graph is not Eulerian. [1]

(ii) Describe how the graph can be made Eulerian by adding one more arc. [1]

(d) The vertices of the graph K_3 are connected to the vertices of the graph K_4 to form the graph K_7.

Explain why 12 arcs are needed connecting K_3 to K_4. [3]

18(a) Alfie has a set of 15 cards numbered consecutively from 1 to 15.

He chooses two of the cards.

How many different sets of two cards are possible? [2]

(b) Alfie places the two cards side by side to form a number with 2, 3 or 4 digits.

Explain why there are fewer than $^{15}\mathrm{P}_2 = 210$ possible numbers that can be made. [1]

(c) Explain why, with these cards, 1 is the lead digit more often than any other digit. [1]

(d) Alfie makes the number 113, which is a 3-digit prime number.

Alfie says that the problem of working out how many 3-digit prime numbers can be made using two of the cards is a construction problem, because he is trying to find all of them.

Explain why Alfie is wrong to say this is a construction problem. [1]

19 Sheona and Tim are making a short film. The activities involved, their durations and immediate predecessors are given in the table below.

	Activity	Duration (days)	Immediate predecessors	S	T
A	Planning	2	–	✓	✓
B	Write script	1	A	✓	
C	Choose locations	1	A		✓
D	Casting	0.5	A	✓	
E	Rehearsals	2	B, D	✓	
F	Get permissions	1	C		✓
G	First day filming	1	E, F	✓	
H	First day edits	1	G		✓
I	Second day filming	0.5	G	✓	
J	Second day edits	2	H, I		✓
K	Finishing	1	J	✓	✓

(a) By using an activity network, find:

- the minimum project completion time
- the critical activities
- the float on each non-critical activity. [7]

(b) Give two reasons why the filming may take longer than the minimum project completion time. [2]

Each activity will involve either Sheona or Tim or both.

- The activities that Sheona will do are ticked in the S column.
- The activities that Tim will do are ticked in the T column.
- They will do the planning and finishing together.
- Some of the activities involve other people as well.

An additional restriction is that Sheona and Tim can each only do one activity at a time.

(c) Explain why the minimum project completion is longer than in part (a) when this additional restriction is taken into account. [2]

(d) The project must be completed in 14 days. Find:

(i) the longest break that either Sheona or Tim can take, [2]

(ii) the longest break that Sheona and Tim can take together, [2]

(iii) the float on each activity. [2]

20 Greetings cards are sold in luxury, standard and economy packs.

The table shows the cost of each pack and number of cards of each kind in the pack.

Pack	Cost (£)	Handmade cards	Cards with flowers	Cards with animals	Other cards	Total number of cards
Luxury	6.50	10	5	5	0	20
Standard	5.00	5	10	5	10	30
Economy	4.00	0	10	10	20	40

Alice needs 25 cards, of which at least 8 must be handmade cards, at least 8 must be cards with flowers and at least 4 must be cards with animals.

(a) Explain why Alice will need to buy at least two packs of cards. [2]

Alice does not want to spend more than £12 on the cards.

(b) (i) List the combinations of packs that satisfy all Alice's requirements. [2]

(ii) Which of these is the cheapest? [1]

Ben offers to buy any cards that Alice buys but does not need. He will pay 12 pence for each handmade card and 5 pence for any other card.

Alice does not want her net expenditure (the amount she spends minus the amount that Ben pays her) on the cards to be more than £12.

(c) Show that Alice could now buy two luxury packs. [2]

Alice decides to buy exactly 2 packs, of which x are luxury packs, y are standard packs and the rest are economy packs.

(d) Give an expression, in terms of x and y only, for the number of cards of each type that Alice buys. [4]

Alice wants to minimise her net expenditure.

(e) Find, and simplify, an expression for Alice's minimum net expenditure in pence, in terms of x and y. You may assume that Alice buys enough cards to satisfy her own requirements. [3]

(f) Find Alice's minimum net expenditure [2]

21 The complete bipartite graph $K_{3,4}$ connects the vertices {2, 4, 6} to the vertices {1, 3, 5, 7}.

(a) How many arcs does the graph $K_{3,4}$ have? [1]

(b) Deduce how many different paths are there that pass through each of the vertices once and once only. The direction of travel of the path does not matter. [3]

The arcs are weighted with the product of the numbers at the vertices that they join.

(c) (i) Use an appropriate algorithm to find a minimum spanning tree for this network. [3]

(ii) Give the weight of the minimum spanning tree. [1]

22 In the pay-off matrix below, the entry in each cell is of the form (r, c), where r is the pay-off for the player on rows and c is the pay-off for the player on columns when they play that cell.

	P	Q	R
X	(1, 4)	(5, 3)	(2, 6)
Y	(5, 2)	(1, 3)	(0, 1)
Z	(4, 3)	(3, 1)	(2, 1)

(a) Show that the play-safe strategy for the player on columns is P. [2]

(b) Demonstrate that the game is not stable. [2]

The pay-off for the cell in row Y, column P is changed from (5, 2) to (y, p), where y and p are real numbers.

(c) What is the largest set of values A, so that if $y \in A$ then row Y is dominated by another row? [1]

(d) Explain why column P can never be redundant because of dominance. [1]

23 Mo eats exactly 6 doughnuts in 4 days.

(a) What does the pigeonhole principle tell you about the number of doughnuts Mo eats in a day? [1]

Mo eats exactly 6 doughnuts in 4 days, eating at least 1 doughnut each day.

(b) Show that there must be either two consecutive days or three consecutive days on which Mo eats a total of exactly 4 doughnuts. [3]

Mo eats exactly 3 identical jam doughnuts and exactly 3 identical iced doughnuts over the 4 days.

The number of jam doughnuts eaten on the four days is recorded as a list, for example 1, 0, 2, 0. The number of iced doughnuts eaten is not recorded.

(c) Show that 20 different such lists are possible. [3]

24 Some jars need to be packed into small crates.

There are 17 small jars, 7 medium jars and 3 large jars to be packed.

- A medium jar takes up the same space as four small jars.
- A large jar takes up the same space as nine small jars.

Each crate can hold:

- at most 12 small jars,
- or at most 3 medium jars,
- or at most 1 large jar (and 3 small jars),
- or a mixture of jars of different sizes.

(a) One strategy is to fill as many crates as possible with small jars first, then continue using the medium jars and finally the large jars.
Show that this method will use seven crates. [2]

The jars can be packed using fewer than seven crates.
(b) The jars are to be packed in the minimum number of crates possible.

- Describe how the jars can be packed in the minimum number of crates.
- Explain how you know that this is the minimum number of crates. [3]

Some other numbers of the small, medium and large jars need to be packed into boxes.

The number of jars that a box can hold is the same as for a crate, except that

- a box cannot hold 3 medium jars.

(c) Describe a packing strategy that will minimise the number of boxes needed. [1]

25 An online magazine consists of an editorial, articles, reviews and advertisements.

The magazine must have a total of at least 12 pages. The editorial always takes up exactly half a page. There must be at least 3 pages of articles and at most 1.5 pages of reviews. At least a quarter but fewer than half of the pages in the magazine must be used for advertisements.

Let x be the number of pages used for articles, y be the number of pages used for reviews and z be the number of pages used for advertisements.

The constraints on the values of x, y and z are:

$x + y + z \geq 11.5$

$x \geq 3$

$y \leq 1.5$

$2x + 2y - 2z + 1 \geq 0$

$2x + 2y - 6z + 1 \leq 0$

$y \geq 0$

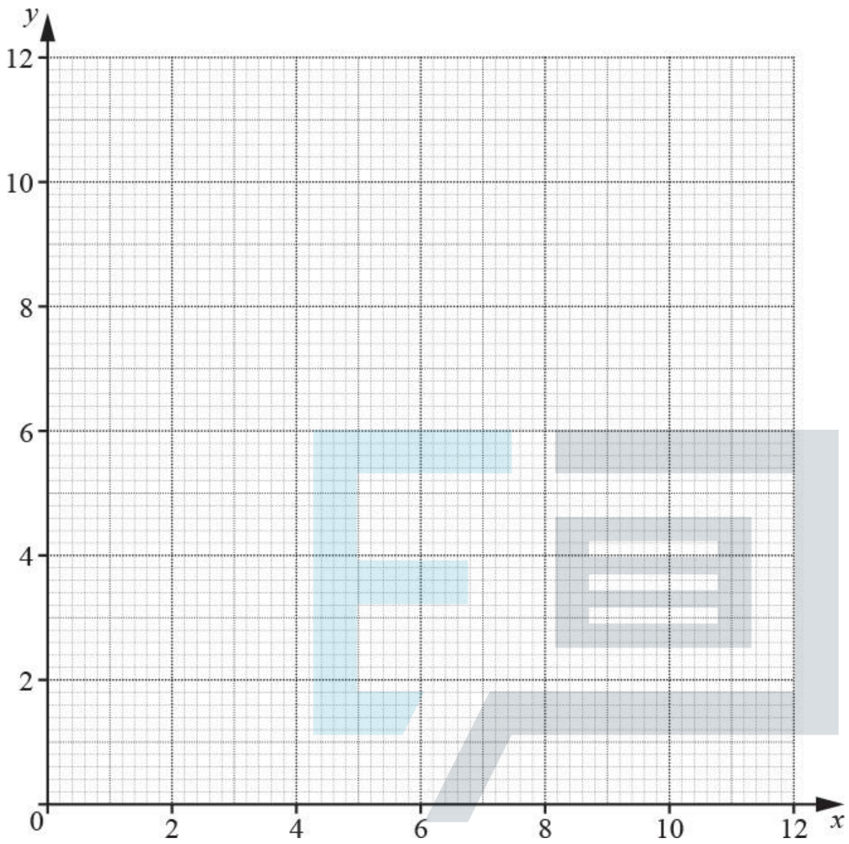
(a) (i) Explain why $x + y + z \geq 11.5$. [1]

(ii) Explain why only one non-negativity constraint is needed. [1]

(iii) Show that the requirement that at least one quarter of the pages in the magazine must be used for advertisements leads to the constraint $2x + 2y - 6z + 1 \leq 0$. [2]

Advertisements bring in money but are not popular with the subscribers. The editor decides to limit the number of pages of advertisements to at most four.

(b) Graph the feasible region in the case when z = 4 using the axes below. [5]



To be successful the magazine needs to maximise the number of subscribers.
The editor has found that when $z \le 4$ the expected number of subscribers is given by $P = 300x + 400y$.

(c) (i) What is the maximum expected number of subscribers when $z = 4$? [2]

(ii) By first considering the feasible region for $z = k$, where $k \le 4$, find an expression for the maximum number of subscribers in terms of k. [5]

26 (a) How many arcs does the complete bipartite graph $K_{5,5}$ have? [1]

A subgraph of $K_{5,5}$ contains 5 arcs joining each of the elements of the set {1, 2, 3, 4, 5} to an element in a permutation of the set {1, 2, 3, 4, 5}. Suppose that r is connected to $p(r)$ for r = 1, 2, 3, 4, 5.

(b) How many permutations would have $p(1) \neq 1$? [2]

(c) Using the pigeonhole principle, show that for every permutation of {1, 2, 3, 4, 5}, the product $\Pi^5_{r=1}(r - p(r))$ is even (i.e. an integer multiple of 2, including 0). [3]

(d) Is the result in part (c) true when the permutation is of the set {1, 2, 3, 4, 5, 6}?
Give a reason for your answer. [2]

27 Deva is having some work done on his house. The table shows the activities involved, their durations and their immediate predecessors.

Activity		Immediate predecessors	Duration (hours)
A	Have skip delivered	–	3
B	Remodel walls	A	3
C	Buy new fittings	–	2
D	Fit electrics	B	2
E	Fit plumbing	B	2
F	Install fittings	C, E	3
G	Plastering	D, E	2
H	Decorating	F, G	3

(a) Model this information as an activity network. [3]

(b) Find the minimum time in which the work can be completed. [2]

(c) Describe the effect on the minimum project completion time of each of the following happening individually.

(i) The duration of activity A is increased to 3.5 hours. [1]

(ii) The duration of activity D is increased to 4 hours. [1]

(iii) The duration of activity F is decreased to 2 hours. [1]

The decorators working on activity H cannot work for 3 hours without a break.

(d) How would you adapt your model to incorporate the break? [1]

28 Lee and Maria are playing a strategy game. The tables below show the points scored by Lee and the points scored by Maria for each combination of strategies.

Points scored by Lee

Lee's choice	Maria's choice W	X	Y	Z
P	5	8	3	4
Q	4	2	7	5
R	2	1	5	3

Points scored by Maria

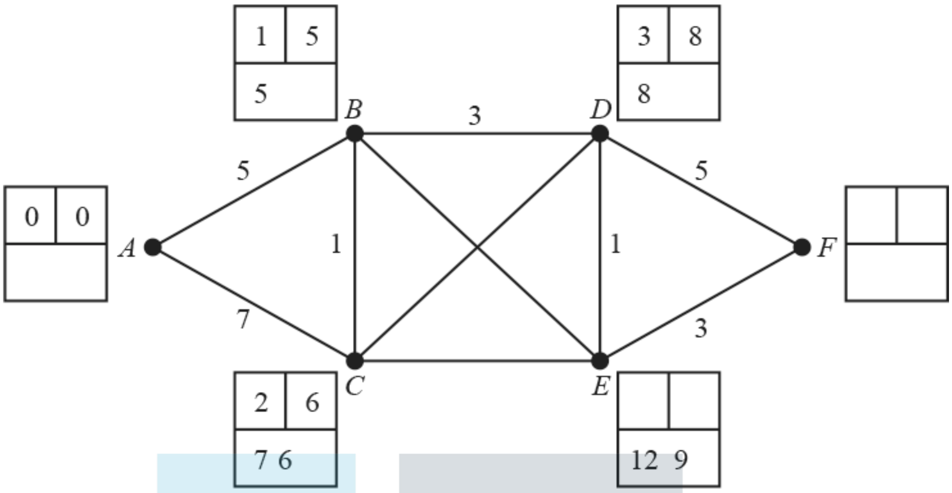
Lee's choice	Maria's choice W	X	Y	Z
P	3	0	5	4
Q	4	6	1	3
R	6	7	3	5

(a) Show how this game can be reformulated as a zero-sum game. [2]

	W	X	Y	Z
P				
Q				
R				

(b) By first using dominance to eliminate one of Lee's choices, use a graphical method to find the optimal mixed strategy for Lee. [7]

29 The diagram shows an incomplete solution to the problem of using Dijkstra's algorithm to find a shortest path from A to F. Any cell that has values in it is complete.



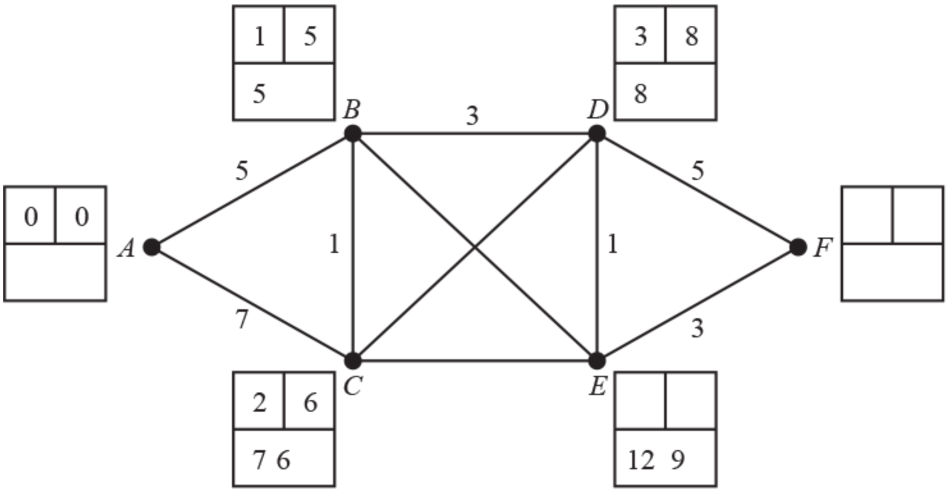
(a) (i) Find the missing weight on arc BE. [1]

(ii) What can you deduce about the missing weight on arc CD? [2]

You are now given that the weight of arc CE is not 3.

(b) (i) What can you deduce about the missing weight on arc CE? [1]

(ii) Complete the labelling of the boxes at *E* and *F* on the diagram below. [2]



Suppose that there are two shortest routes from *A* to *F*.

(c) Show how trace back is used to find the shortest routes from *A* to *F*. [3]

30 (a) (i) Show that the number of arrangements of 25 distinct objects is an integer multiple of 5^6. [3]

(ii) Explain how this shows that the number of arrangements of 25 distinct objects is a whole number of millions. [1]

(b) (i) Calculate the values of [2]

- INT(720 ÷ 25)
- INT(720 ÷ 125).

(ii) Deduce the largest power of 10 that is a factor of 720! [2]

(c) Use the inclusion-exclusion principle to find the number of integers from 1 to 720 that are not divisible by either 2 or 5. [2]

31 Each day a potter makes dishes, mugs and pots.

Let x = the number of dishes
y = the number of mugs
z = the number of pots
that are made each day.

The dishes, mugs and pots have to be baked together in a kiln which is only used once each day.

There is enough space in the kiln for 15 dishes, with no mugs or pots, or 50 mugs, with no dishes or pots, or 200 pots, with no dishes or mugs. Usually the kiln holds a mixture of dishes, mugs and pots.

(i) Show that this leads to a constraint of the form $ax + by + cz \le d$, where a, b, c and d are integers to be determined. [3]

Other constraints come from the amount of clay required, the potter's time and customer demand.

The potter wants to maximise the profit, £P, from selling the dishes, mugs and pots.

This leads to a linear programming problem, which is represented as a Simplex tableau with four slack variables, s, t, u and v. After one iteration the resulting tableau is:

P	x	y	z	s	t	u	v	RHS
1	0	0.5	–0.75	0	0	0	1.25	50
0	0	0	–0.5	1	0	0	–2.5	20
0	0	0.5	0.25	0	1	0	–0.75	6
0	0	–8	–7	0	0	1	–10	200
0	1	0.5	0.25	0	0	0	0.25	10

(ii) Which column was the first pivot chosen from? [1]

(iii) Carry out a second iteration. [4]

P	x	y	z	s	t	u	v	RHS

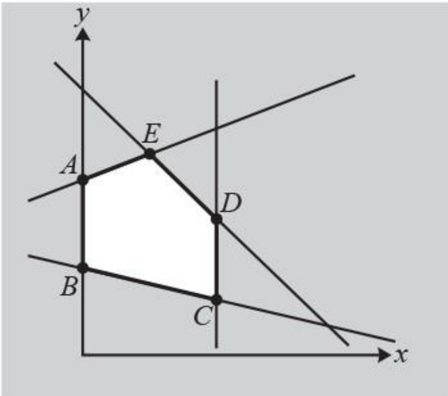
The final tableau is:

P	x	y	z	s	t	u	v	RHS
1	1	2	0	0	2	0	0	72
0	4	1	0	1	−2	0	0	48
0	3	2	1	0	1	0	0	36
0	31	6	0	0	−3	1	0	492
0	1	0	0	0	−1	0	1	4

(iv) (a) Use this tableau to state the number of dishes, mugs and pots that should be made each day and the potter's profit. [2]

(b) If the potter makes this number of dishes, mugs and pots, why might the profit be less than the amount given in the answer to part (iv)(a)? [1]

32 The graph below shows the feasible region (unshaded plus boundaries) for the linear programming (LP) problem below. The vertices of the feasible region are marked as *A*, *B*, *C*, *D* and *E*.



LP problem

Minimise $3x + 13y$

Subject to $x \leqslant 4$

$x + 4y \geqslant 8$

$-x + 3y \leqslant 12$

$2x + y \leqslant 11$

$x \geqslant 0,\ y \geqslant 0$

(i) Find the coordinates of the vertices *A*, *B*, *C*, *D* and *E*. [4]

The objective of the problem is to minimise $3x + 13y$.

(ii) Find the minimum feasible value of $3x + 13y$. [2]

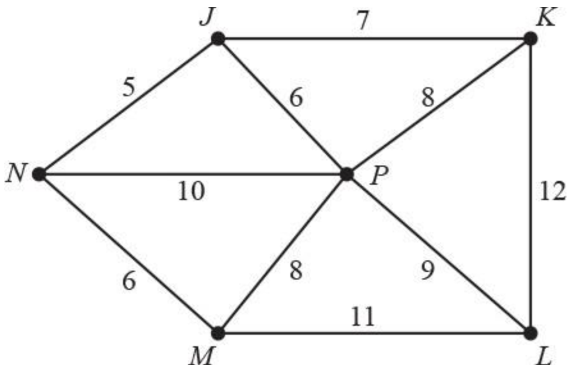
The constraint $x \leq 4$ is removed, but otherwise the problem is unchanged.

(iii) Find the minimum feasible value of $3x + 13y$ for the new problem. [3]

Suppose now that there is an additional restriction that x and y must be integers.

(iv) Show that (5, 1) is feasible but does not give the minimum feasible value of $3x + 13y$ for the new problem with the additional restriction. [2]

33 The network below represents the road routes between six villages. The weights on the arcs represent distances in miles.

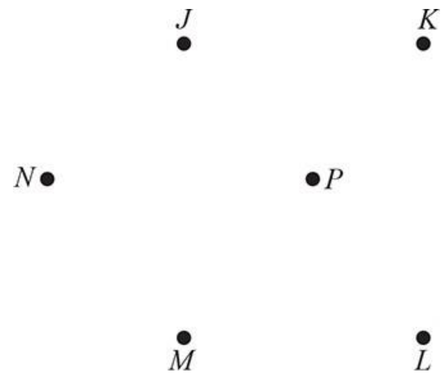


Phoebe needs to deliver parcels to each village. She lives in village P. She wants to find a route that starts at P, visits each of the other villages (in some order) and finishes at P. She wants her route to be as short as possible.

(i) Use the nearest neighbour method to find a suitable route that starts at P, visits each of the other villages (in some order) and finishes at P. Write down your route and its length in miles. [2]

(ii) Find a route that passes through every vertex, starting and ending at P, that is shorter than the route found in part (i). Write down your route and its length in miles. [2]

(iii) Use Kruskal's algorithm to find a minimum spanning tree for the network. You should show the order in which you consider the arcs. Draw your tree and give its total weight. [4]



The road joining L to M is damaged. It is replaced by a new road that joins L to M of length k miles.

The new road is not part of any minimum spanning tree for the new network.

(iv) Find the possible values of k, explaining your reasoning. [2]

Although not being part of any minimum spanning tree for the new network, the new road is part of the shortest route through every vertex, starting and ending at P.

(v) Find the possible values of k, explaining your reasoning. [3]

34 The table below shows the travel times (in minutes) between eight railway stations, A to H. Only travel times for direct routes (that do not involve changing trains) are shown, although an indirect route (changing trains) may be quicker. The travel times for seventeen direct routes are shown.

For this question you must work directly from the values in the table and not from a diagram.

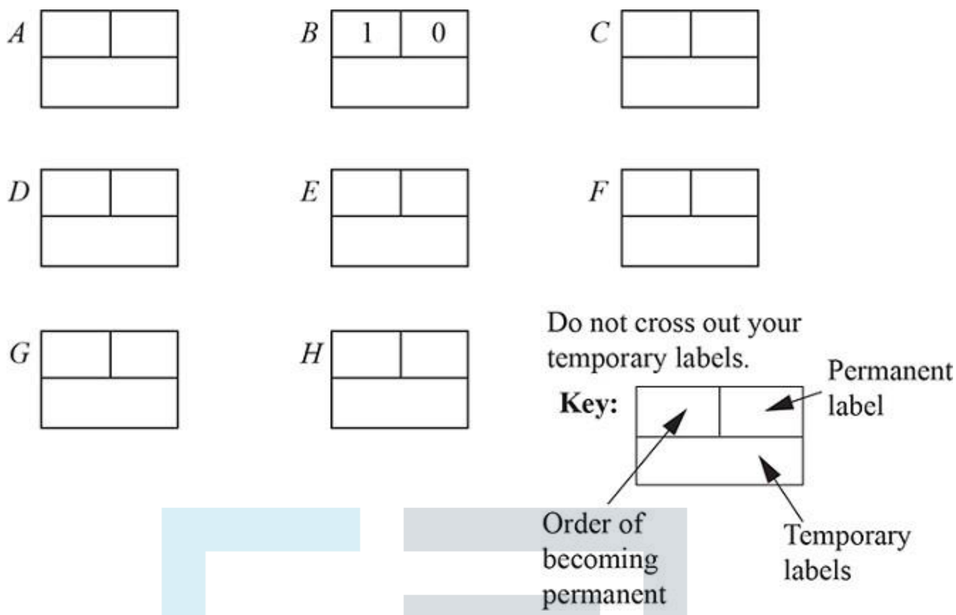
	A	B	C	D	E	F	G	H
A	–	6	22	13	27	–	14	12
B	6	–	19	7	–	–	–	–
C	22	19	–	25	5	–	8	10
D	13	7	25	–	–	16	–	–
E	27	–	5	–	–	8	13	15
F	–	–	–	16	8	–	–	–
G	14	–	8	–	13	–	–	2
H	12	–	10	–	15	–	2	–

The sum of all the values in the table is 444.

(i) Using only the direct routes shown in the table, and without drawing the network, list the stations that are connected to an odd number of other stations (the odd vertices). [1]

(ii) Find the shortest travel time from A to F that only involves changing trains once. [2]

(iii) Without drawing the network, use Dijkstra's algorithm to find the shortest travel times (direct or indirect) from B to each of the other stations. [4]



(iv) Deduce the shortest travel time for a closed route (starting and finishing at A) that involves travelling each of the seventeen direct routes shown in the table. Write down the arcs that are repeated and find how many times station C is travelled through when using this route. [4]

35 The following algorithm displays three values for each positive integer input.

Line 10	Input a positive integer, N
Line 20	If N is an odd number jump to line 80
Line 30	Let $P = 1$
Line 40	Let $M = N$
Line 50	Divide the value of M by 2 and make this the new value of M
Line 60	If M is an odd number jump to line 90
Line 70	Increase P by 1 and go back to line 50
Line 80	Let $P = 0$ and let $M = N$
Line 90	Let $A = (M^2 + 1) \times 2^{(P-1)}$
Line 100	Let $B = A - 2^P$
Line 110	Display N, B, A and stop

(i) Apply the algorithm to the input $N = 5$. You only need to write down values when they change. You should record the line numbers but there is no need to record passing through lines 20 and 60. [4]

Line	N	P	M	A	B	Display (N, B, A)
10						

(ii) Apply the algorithm to the input N = 20. [5]

Line	N	P	M	A	B	Display (N, B, A)
10						

(iii) Explain what happens when $N = 2^x$, for some positive integer x. [4]

36 A *simple* graph is one in which any two vertices are directly connected by at most one arc and no vertex is directly connected to itself. A *connected* graph is one in which every vertex is joined, directly or indirectly, to every other vertex. A *simply connected* graph is one that is both simple and connected.

Josh has drawn a simply connected graph that has exactly six vertices and exactly eight arcs. Four of the vertex orders are 2, 2, 4 and 5.

(i) Without doing any calculations and without drawing any graphs, how do the given vertex orders show that:

(a) Josh's graph is not Eulerian, [1]

(b) Josh's graph is semi-Eulerian. [1]

(ii) Without drawing any examples, calculate the values of the two missing vertex orders. Briefly explain your reasoning, by referring to properties of the graph and their consequences on the vertex orders. [2]

(iii) Argue in words, without drawing any examples, why any graph that fits these requirements has the same structure (in the sense of which vertices are directly connected to one another) as the one that Josh has drawn. [3]

37 The following list is to be arranged into increasing order, from smallest to largest.

12	10	3	6	12	4	5

(i) Use bubble sort to arrange the list into increasing order starting at the left-hand side of the list. Record the whole list at the end of each pass. There is no need to show the individual comparisons or swaps. [3]

A different list was arranged into increasing order using bubble sort. The following list is what was obtained after the first pass.

27	8	14	40

(ii) (a) Write down two possible orders for the original list. [2]

(b) How many more passes will be needed to complete the sort (excluding the first)? [1]

38 A sweet shop sells three different types of boxes of chocolate truffles. The cost of each type of box and the number of truffles of each variety in each type of box are given in the table below.

Type	Cost (£)	Milk chocolate	Plain chocolate	White chocolate	Nutty chocolate
Assorted	2.00	5	5	5	5
No Nuts	1.00	5	8	7	0
Speciality	2.50	5	4	9	2

Narendra wants to buy some boxes of truffles so that in total he has at least 20 milk chocolate, 10 plain chocolate, 16 white chocolate and 12 nutty chocolate truffles.

(a) Explain why Narendra needs to buy at least four boxes of truffles. [1]

(b) Narendra decides that he will buy exactly four boxes. Determine the minimum number of Assorted boxes that Narendra must buy. [1]

(c) For your answer in part (b),

- list all the feasible solutions and
- find the cheapest solution. [3]

Narendra finds that the sweet shop has sold out of Assorted boxes, but he then spots that it also sells small boxes of milk chocolate truffles and small boxes of nutty chocolate truffles. Each small box contains 4 truffles (all of one variety) and costs £0.50.

He decides to buy x boxes of No Nuts and y boxes of Speciality, where $x + y < 4$, so that he has at least 10 plain chocolate and 16 white chocolate truffles. He will then buy as many small boxes as he needs to give a total of at least 20 milk chocolate and 12 nutty chocolate truffles.

(d) (i) Set up constraints on the values of x and y. [2]

(ii) Represent the feasible region graphically. [2]

(iii) Hence determine the cheapest cost for Narendra. [3]

39 A complete graph on five vertices is weighted to form a network, as given in the weighted matrix below.

	A	B	C	D	E
A	-	9	5	4	2
B	9	-	7	5	7
C	5	7	-	6	8
D	4	5	6	-	5
E	2	7	8	5	-

(a) Apply Prim's algorithm to the weighted matrix above to construct a minimum spanning tree for the five vertices.

Draw your minimum spanning tree, stating the order in which you built the tree and giving its total weight.

[4]

(b) (i) Using only the arcs in the minimum spanning tree, which vertex should be chosen to find the smallest total of the weights of the paths from that vertex to each of the other vertices? [1]

(ii) State the minimum total for this vertex. [1]

(c) Show that the total number of comparisons needed to find a minimum spanning tree for a 5×5 matrix is 16. [3]

(d) If a computer takes 4 seconds to find a minimum spanning tree for a network with 100 vertices, how long would it take to find a minimum spanning tree for a network with 500 vertices? [2]

40 The following masses, in kg, are to be packed into bins.

8	5	9	7	7	9	1	3	3	8

(a) Chloe says that first-fit decreasing gives a packing that requires 4 bins, but first-fit only requires 3 bins. Find the maximum capacity of the bins. [6]

First-fit requires one pass through the list and the time taken may be regarded as being proportional to the length of the list. Suppose that shuttle sort was used to sort the list into decreasing order.

(b) What can be deduced, in this case, about the order of the time complexity, T(n), for first-fit decreasing? [2]

41 There are three non-isomorphic trees on five vertices.

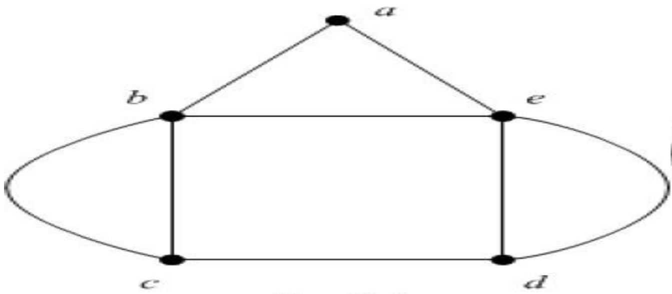
(a) Draw an example of each of these trees. [1]

(b) State three properties that must be satisfied by the vertex orders of a tree on six vertices. [3]

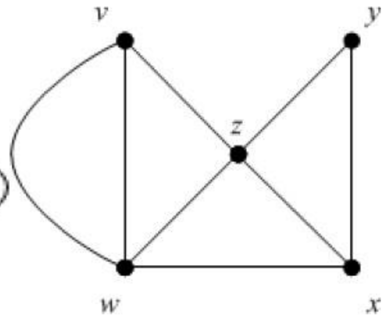
(c) List the five different sets of possible vertex orders for trees on six vertices. [2]

(d) Draw an example of each type listed in part (c). [2]

42 Two graphs are shown below. Each has exactly five vertices with vertex orders 2, 3, 3, 4, 4.



Graph 1



Graph 2

(a) Write down a semi-Eulerian route for graph 1. [1]

(b) Explain how the vertex orders show that graph 2 is also semi-Eulerian. [1]

(c) By referring to specific vertices, explain how you know that these graphs are not simple. [2]

(d) By referring to specific vertices, explain how you know that these graphs are not isomorphic. [2]

43 A zero-sum game is being played between two players, X and Y. The pay-off matrix for X is given below.

		Player Y	
		Strategy R	Strategy S
	Strategy P	4	–2
Player X	Strategy Q	–3	1

(a) Find an optimal mixed strategy for player X. [5]

(b) Give one assumption that must be made about the behaviour of Y in order to make the mixed strategy of Player X valid. [1]

44 Some of the activities that may be involved in making a cup of tea are listed below.

A: Boil water.
B: Put teabag in teapot, pour on boiled water and let tea brew.
C: Get cup from cupboard.
D: Pour tea into cup.
E: Add milk to cup.
F: Add sugar to cup.

Activity A must happen before activity B.
Activities B and C must happen before activity D.
Activities E and F cannot happen until after activity C.
Other than that, the activities can happen in any order.

(a) Lisa does not take milk or sugar in her tea, so she only needs to use activities A, B, C and D. In how many different orders can activities A, B, C and D be arranged, subject to the restrictions above? [1]

(b) Mick takes milk but no sugar, so he needs to use activities A, B, C, D and E. Explain carefully why there are exactly nine different orders for these activities, subject to the restrictions above. [3]

(c) Find the number of different orders for all six activities, subject to the restrictions above. Explain your reasoning carefully. [3]

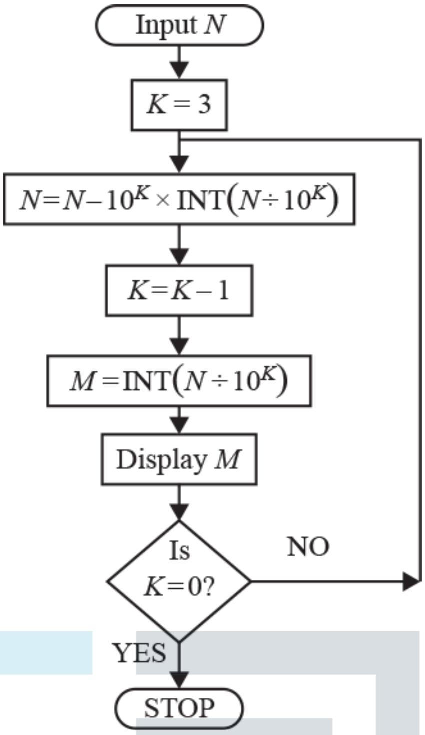
45 Hussain wants to travel by train from Edinburgh to Southampton, leaving Edinburgh after 9 am and arriving in Southampton by 4 pm. He wants to leave Edinburgh as late as possible.

Hussain rings the train company to find out about the train times. Write down a question he might ask that leads to

(A) an existence problem,

(B) an optimisation problem. [2]

46



(a) Trace through the flowchart above using the input N = 97 531. You only need to record values when they change. [3]

N	K	M	Display

(b) Explain why the process in the flowchart is finite. [1]

47 Rahul is decorating a room. He needs to decorate at least $30m^2$ of the walls using paint, panelling or wallpaper.

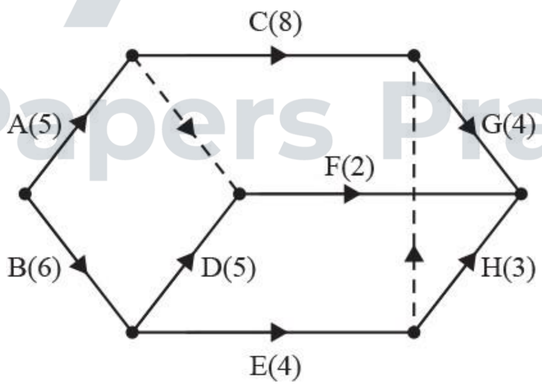
The cost (£) and the time required (hours) to decorate 1 m^2 of wall are shown in the table.

	Cost	Time
Paint	1.12	0.50
Panelling	4.62	0.34
Wallpaper	1.61	0.28

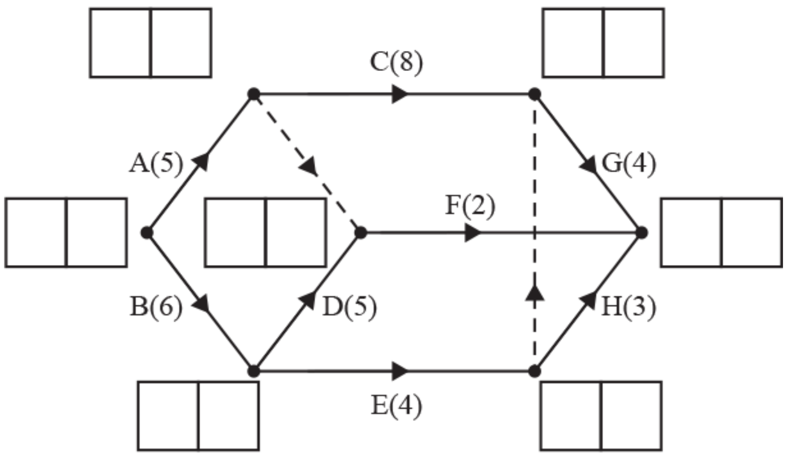
Rahul wants to complete decorating the walls in no more than 8 hours and wants to minimise the cost.

Set up an LP formulation for Rahul's problem, defining your variables. You are not required to solve this LP problem. [6]

48 The activities involved in a project and their durations are represented in the activity network below.



(a) Carry out a forward pass and a backward pass through the network. [3]



(b) Find the float for each activity. [3]

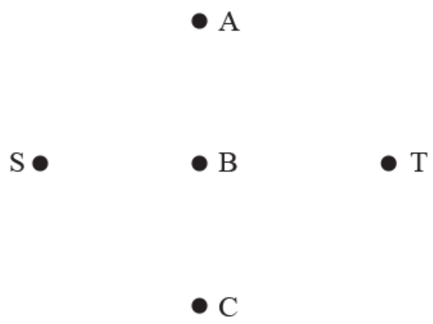
Activity	A	B	C	D	E	F	G	H
Float								

A delay means that the duration of activity E increases to x.

(c) Find the values of x for which activity E is not a critical activity. [2]

49 Tom is planning a day out walking. He wants to start from his sister's house (S), then visit three places A, B and C (once each, in any order) and then finish at his own house (T).

(a) Complete the graph given below showing all possible arcs that could be used to plan Tom's walk. [1]



Tom needs to keep the total distance that he walks to a minimum, so he weights his graph.

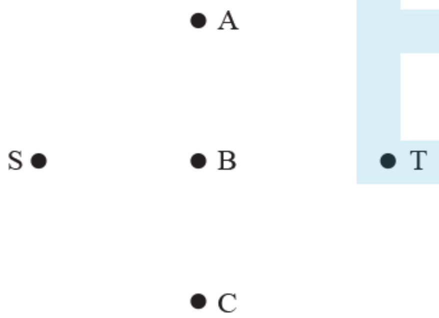
(b) (i) Why would finding the shortest path from S to T, on Tom's network, not necessarily solve Tom's problem? [1]

(ii) Why would finding a minimum spanning tree, on Tom's network, not necessarily solve Tom's problem? [1]

The distance matrix below shows the direct distances, in km, between places.

	S	A	B	C	T
S	0	3	2	5	4
A	3	0	2	2.5	2
B	2	2	0	3.2	2.5
C	5	2.5	3.2	0	2
T	4	2	2.5	2	0

(c) • Use an appropriate algorithm to find a minimum spanning tree for Tom's network.
• Give the total weight of the minimum spanning tree. [3]



Total weight ____________________

(d) • Find the route that solves Tom's problem.
• Give the minimum distance that Tom must walk. [2]

Route:

Minimum distance:

50 In each round of a card game two players each have four cards. Every card has a coloured number.

- Player A's cards are red 1, blue 2, red 3 and blue 4.
- Player B's cards are red 1, red 2, blue 3 and blue 4.

Each player chooses one of their cards. The players then show their choices simultaneously and deduce how many points they have won or lost as follows:

- If the numbers are the same both players score 0.
- If the numbers are different but are the same colour, the player with the lower value card scores the product of the numbers on the cards.
- If the numbers are different and are different colours, the player with the higher value card scores the sum of the numbers on the cards.
- The game is zero-sum.

(a) Complete the pay-off matrix for this game, with player A on rows. [3]

		B				
		red 1	red 2	blue 3	blue 4	
	red 1	0				
	blue 2		0			
A	red 3			0		
	blue 4				0	

(b) Determine the play-safe strategy for each player. [2]

Play-safe for A is

Play-safe for B is

(c) Use dominance to show that player A should not choose red 3. You do not need to identify other rows or columns that are dominated. [1]

(d) Determine, with a reason, whether the game is stable or unstable. [1]

51 This list is to be sorted into decreasing order, ending up with 31 in the first position and 4 in the last position.

15 4 12 23 9 14 16 27 31

Initially bubble sort is used.

(a) Record the list at the end of the first, second and third passes. You do not need to show the individual swaps in each pass. [2]

	15	4	12	23	9	14	16	27	31
After 1st pass									
After 2nd pass									
After 3rd pass									

After the fourth pass the list is:

23 15 16 27 31 14 12 9 4

The sorting is then continued using shuttle sort on this list.

(b) Record the list at the end of each of the first, second and third passes of shuttle sort. You do not need to show the individual swaps in each pass. [2]

	23	15	16	27	31	14	12	9	4
After 1st pass									
After 2nd pass									
After 3rd pass									

(c) How many passes through shuttle sort are needed to complete the sort? Explain your reasoning. [2]

52 (a) A digraph is represented by the adjacency matrix below.

$$\begin{array}{cc} & \text{to} \\ & \begin{array}{ccc} A & B & C \end{array} \\ \text{from} \begin{array}{c} A \\ B \\ C \end{array} & \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \end{array}$$

(i) For each vertex, write down

- the indegree,
- the outdegree. [2]

Indegrees: A= B= C=

Outdegrees: A= B= C=

(ii) Explain how the indegrees and outdegrees show that the digraph is semi-Eulerian. [2]

(b) A graph is represented by the adjacency matrix below.

$$\begin{array}{cc} & \begin{array}{ccc} D & E & F \end{array} \\ \begin{array}{c} D \\ E \\ F \end{array} & \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix} \end{array}$$

(i) Explain how the numerical entries in the matrix show that this can be drawn as an undirected graph. [1]

(ii) Explain how the adjacency matrix shows that this graph is semi-Eulerian. [2]

(c) By referring to the vertex degrees, explain why the loop from A to itself is shown as a 1 in the adjacency matrix in part (a) but the loop from D to itself is shown as a 2 in the adjacency matrix in part (b). [2]

53 (a) List the 15 partitions of the set {A, B, C, D}. [3]

Amy, Bao, Cal and Dana want to travel by taxi from a meeting to the railway station. They book taxis but only two taxis turn up. Each taxi must have a minimum of one passenger and can carry a maximum of four passengers. Dana jumps into one of the taxis.

(b) Find the number of ways that Amy, Bao and Cal can be split between the two taxis. [1]

Amy does not want to travel in the same taxi as Bao.

(c) Determine the number of ways that Amy, Bao and Cal can be split between taxis with this additional restriction. [2]

Six people want to travel by taxi from a hotel to the railway station using taxis. There must be a minimum of one passenger and a maximum of four passengers in each taxi. The taxis may be regarded as being indistinguishable.

(d) Calculate how many ways there are of splitting the six people between taxis. [6]

END OF QUESTION PAPER